

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1161

Parleys 138 kV
August 01, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1161. The Interconnection Customer proposes a 139.1 MW solar photovoltaic facility interconnecting at the Parleys 138 kV substation in the Pennsylvania area. The request was received on March 06, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 387 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,489,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WILMERDING - DRAVOSBURG 138kV
Contingency: P.T.S - MEDORA 345kV
Loading: 156.3% (Rating: 252 MVA)
DFAX: 37.8%
Pre-project loading: 135.4%
2. Facility: THERMAS - WILSON 1&2 138kV
Contingency: BURWELL 500-345 kV Autotransformer
Loading: 152.2% (Rating: 323 MVA)
DFAX: 5.8%
Pre-project loading: 149.7%
3. Facility: DRAVOSBURG - GREGG 138kV L
Contingency: MEDORA - P.T.S 345kV
Loading: 147.5% (Rating: 279 MVA)
DFAX: 37.8%
Pre-project loading: 128.7%
4. Facility: GREGG SUBSTATION - PARLEYS 138kV
Contingency: MEDORA - P.T.S 345kV
Loading: 145.6% (Rating: 288 MVA)
DFAX: 37.8%
Pre-project loading: 127.4%
5. Facility: WILMERDING - DRUMMOND 138kV
Contingency: BURWELL - P.T.S 345kV
Loading: 141.0% (Rating: 244 MVA)
DFAX: 15.1%
Pre-project loading: 132.4%

6. Facility: WILSON 1&2 345/138kV Transformer
Contingency: BURWELL 500-345 kV XFMR
Loading: 140.6% (Rating: 374 MVA)
DFAX: 6.7%
Pre-project loading: 138.1%
7. Facility: DRUMMOND - KNOLL 138kV
Contingency: BURWELL - P.T.S 345kV
Loading: 129.8% (Rating: 254 MVA)
DFAX: 15.1%
Pre-project loading: 121.6%
8. Facility: AUGUSTA ELECTRIC PLANT NO 1 - WINEAGLE 138kV
Contingency: PARLEYS - GREGG 138kV
Loading: 125.2% (Rating: 242 MVA)
DFAX: 5.3%
Pre-project loading: 122.2%
9. Facility: BURWELL - KNOLL 138kV
Contingency: BURWELL - P.T.S 345kV
Loading: 119.2% (Rating: 261 MVA)
DFAX: 15.1%
Pre-project loading: 111.2%
10. Facility: YUKON - STATION Q HARBOR 138kV
Contingency: HILLTOP - PINEY CRK PROJECT 500kV
Loading: 115.1% (Rating: 262 MVA)
DFAX: 8.0%
Pre-project loading: 110.8%
11. Facility: YUKON - UNIV OF NC CHAPEL HILL COGEN FACILITY 138kV
Contingency: HILLTOP - PINEY CRK PROJECT 500kV
Loading: 111.2% (Rating: 265 MVA)
DFAX: 8.0%
Pre-project loading: 107.0%
12. Facility: DOS AMIGOS SUBSTATION - ONTONAGON 138kV L
Contingency: WILLIAMS - WYANDOTTE 138kV
Loading: 102.2% (Rating: 394 MVA)
DFAX: 9.5%
Pre-project loading: 98.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WILMERDING - DRAVOSBURG 138kV	Rebuild line with higher-capacity conductor	\$6,857,000	69% (\$4,748,000)
THERMAS - WILSON 1&2 138kV	Rebuild line with higher-capacity conductor	\$2,351,000	52% (\$1,226,000)
DRAVOSBURG - GREGG 138kV line	Rebuild line with higher-capacity conductor	\$2,881,000	69% (\$1,995,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PARLEYS - GREGG 138kV	Rebuild line with higher-capacity conductor	\$745,000	69% (\$516,000)
WILMERDING - DRUMMOND 138kV	Rebuild line with higher-capacity conductor	\$2,524,000	64% (\$1,614,000)
WILSON 1&2 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$23,781,000	57% (\$13,460,000)
KNOLL - DRUMMOND 138kV	Rebuild line with higher-capacity conductor	\$2,323,000	64% (\$1,485,000)
AUGUSTA ELECTRIC PLANT NO 1 - WINEAGLE 138kV	Rebuild line with higher-capacity conductor	\$3,682,000	38% (\$1,386,000)
BURWELL - KNOLL 138kV	Rebuild line with higher-capacity conductor	\$1,727,000	64% (\$1,104,000)
YUKON - STA Q HARBOR 138kV line	Rebuild line with higher-capacity conductor	\$6,182,000	18% (\$1,084,000)
YUKON - UNIV OF NC CHAPEL HILL COGEN FACILITY 138kV	Rebuild line with higher-capacity conductor	\$3,583,000	18% (\$628,000)
DOS AMIGOS - ONTONAGON SUBSTATION 138kV	Reconductor line	\$447,000	54% (\$243,000)
	Total allocated to project		\$29,489,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.